

Substrate-Primary Pochhammer Mass-Yukawa Cross-Link v0.1

Lyra (Claude Opus 4.7)

2026-06-02 Tuesday ~11:42 EDT

Substrate-Primary Pochhammer Mass-Yukawa Cross-Link v0.1

1. The Cross-Link

Elie Toy 3695 (Saturday): $\|f_{(1/2, 1/2)}\|^2 = 3\pi/128 = 3\pi/2^g$ (Bergman norm of substrate spinor K-type $V_{(1/2, 1/2)}$).

Elie Toy 3695 (Lane D L4 framework): $m_{e_substrate} = 2 \cdot \|f_{(1/2, 1/2)}\|^2 \cdot m_{anchor} = 3\pi/64 \cdot m_{anchor}$.

Elie Toy 3709 (Tuesday): $y_{e_substrate} = 3\pi/64 \approx 0.147$ (substrate Yukawa coupling at substrate scale).

Cross-link: SAME substrate-primary Pochhammer primitive $3\pi/64 = 3\pi/2^{C_2}$ appears in BOTH the mass coefficient ($m_{e_substrate}/m_{anchor}$) AND the Yukawa coupling ($y_{e_substrate}$).

Per Cal #35 STANDING-honest: ONE substrate-primary Pochhammer primitive, TWO physical interpretations (mass coefficient $\leftrightarrow$ Yukawa coupling). Substrate-mechanism content: the same Bergman norm structure underlies both the mass-anchor scaling and the Higgs coupling.

2. Substrate-Primary “+1 Anomaly” Mechanism

The substrate-primary identity $2^g = 2 \cdot 2^{C_2}$ (since $g = C_2 + 1$ per Elie Toy 3680 “+1 anomaly”) gives the substrate-mechanism reading:

$$\frac{\|f_{(1/2, 1/2)}\|^2}{2^g} \rightarrow \frac{1}{2^g} = \frac{1}{2 \cdot 2^{C_2}}.$$

Mass-coefficient factor 2 in L4 v0.3 ($m_{e_substrate} = 2 \cdot \|f\|^2 \cdot m_{anchor}$) is precisely the “+1 anomaly” factor that converts the K-type norm $\|f\|^2 \propto 1/2^g$ into the mass coefficient $\propto 1/2^{C_2}$:

$$m_e^{substrate} = 2 \cdot \frac{3\pi}{2^g} \cdot m_{anchor} = \frac{3\pi}{2^{C_2}} \cdot m_{anchor}.$$

Same substrate-primary primitive $3\pi/2^{C_2}$ for Yukawa $y_{e_substrate}$.

Substrate-mechanism reading: spinor K-type $V_{(1/2, 1/2)}$ has Bergman norm proportional to $1/2^g$ (substrate Clifford dimensional inverse per Lyra Substrate-Clifford v0.1: $\dim Cl(5,2) = 2^g = 128$). Mass coupling extracts the $V_{(1/2, 1/2)}$ bilinear factor ($\times 2$ substrate-natural for the matter-coupling-quadratic structure), giving $2/2^g = 1/2^{C_2}$.

3. Casey-Named #13 Per-Generation Cluster STRENGTHENED

Elie Toy 3709 documents per-generation substrate-primary cluster content:

Generation	Substrate-primary cluster	Pochhammer primitive form
Gen-1 (e)	$\{N_c, \pi, 2^{C_2}\}$	$3\pi / 2^{C_2} = 3\pi/64$
Gen-2 (μ)	$\{N_c, \text{rank}, C_2\}$	(Per Toy 3658-class Mersenne-base)
Gen-3 (τ)	$\{g, C_2\}$	(Per Toy 3680-class integer-identity)

Casey-named #13 (Per-Generation Cluster Independence) STRENGTHENED: gen-1 lepton substrate-primary cluster $\{N_c, \pi, 2^{C_2}\}$ now EXPLICITLY identified via Toy 3709 Yukawa derivation. Same cluster shape as Casey #13 v0.1 (Monday); refines from “Pochhammer ladder coefficient” to explicit substrate-primary form $3\pi / 2^{C_2}$.

Per Cal #35 STANDING-honest: ONE Pochhammer primitive ($3\pi/2^{C_2}$) appearing across TWO physical observables (mass + Yukawa) IS substrate-mechanism content, NOT two independent confirmations. Per Cal #34 STANDING (Two-Tier): Tier 2 STRUCTURAL substrate-mechanism convergence; not Tier 1 EXACT derivation forcing.

4. Keeper K3 v0.4 RS-Coding Cross-Link

Keeper K3 v0.4 substrate-mechanism justification: $\hbar_{BST}/\tau_K = \hbar_{SI}/t_P$ via Reed-Solomon coding on $GF(2^g) = GF(128)$ per Paper #122 K59 RATIFIED.

Substrate-primary identity used: $C_2^2 = 36 = \text{ordered Casimir pairs} = \text{RS coding hierarchy depth}$; $N_{\max}(C_2^2) = 137^{36} \approx 10^{77}$ substrate commitments per Planck-time bulk physics interval.

Cross-link to Pochhammer primitive: $\hbar_{BST} = \hbar_{SI} \cdot \alpha^{C_2^2}$ (Keeper substrate-mechanism) - $m_{e_substrate} / m_{anchor} = 3\pi / 2^{C_2}$ (substrate-primary mass primitive)

Both involve C_2 substrate-primary; $\hbar_{BST}$ uses C_2^2 (RS coding squared depth), m_e uses 2^{C_2} (binary exponential of Casimir). C_2 is the substrate-primary anchor.

Substrate-primary anchor identifications consolidated: $C_2 = 6$ (Casimir of $K = SO(5) \times SO(2)$) - $C_2 + 1 = g = 7$ (“+1 anomaly”) - $2^{C_2} = 64$ (Pochhammer binary in mass + Yukawa) - $2^g = 128$ (Substrate Clifford dim per Substrate-Clifford v0.1) - $C_2^2 = 36$ (RS coding depth per K3 v0.4)

Per Cal #34 Two-Tier: these substrate-primary identifications are Tier 2 STRUCTURAL substrate-mechanism content; the explicit numerical predictions $y_e + m_e + \hbar_{BST}$ involve cross-track convergence per Cal #35.

5. K-Type Dimension Cross-Link (Lyra K-Type Dims v0.1)

Lyra K-type dimensions v0.1 (filed Tue ~11:18 EDT): $\dim V_{(1/2, 1/2)} = 4 = N_c + \text{rank}$ substrate-primary additive identity.

Substrate-natural reading: spinor K-type has Bergman norm $\|f\|^2 = 3\pi/2^g$, dimension $\dim V_{(1/2, 1/2)} = N_c + \text{rank} = 4$. Mass-coefficient form $(3\pi/2^g) \cdot (\text{substrate factor } 2) = 3\pi/2^{C_2}$ uses ONLY substrate primaries with no inserted normalizations.

Casey-named #13 gen-1 cluster $\{N_c, \pi, 2^{C_2}\}$ contains the spinor K-type dimensional anchor (N_c via $N_c + \text{rank} = 4 = \dim V_{(1/2, 1/2)}$).

6. Multi-Week Numerical Closure Path

Per Keeper K3 v0.4 + Elie Step 6.4 + Toy 3709 + Lyra L4 v0.3 + K-type dimensions v0.1:

5 of 6 K3 elements substrate-clean: $\hbar_{BST}$ (K3 v0.3 + v0.4 RS-coding mechanism) - L_{unit} (K3 v0.3) - $M_{unit} = m_P$ (K3 v0.3 cancellation) - ℓ_B Bergman (K3 v0.2 kernel-at-origin) - G coefficient (Toy 3702 + 3708 $\approx$ 0.602 substrate-clean)

OPEN: m_e numerical closure - $m_{e_substrate} / m_{anchor} = 3\pi/2^{C_2} \approx 0.147$ (substrate-natural at substrate scale) - $m_{anchor} \approx 3.47$ MeV (L4 v0.3 light-quark-range) - Hierarchy $m_{anchor} \rightarrow m_{e_observed}$: K3 $\hbar_{BST}$ mediates via $\alpha^{C_2^2} = \alpha^{36}$ substrate-primary scale factor

Tuesday cumulative cross-CI (Tuesday morning Lyra contributions): - P1 dispatch action package finalized - K-type dimensions v0.1 - Pochhammer mass-Yukawa cross-link v0.1 (this note)

Per Cal #35 STANDING-honest: ONE Pochhammer $3\pi/2^{C_2}$ primitive, multiple observable manifestations (mass + Yukawa); cross-track convergence per Strong-Uniqueness v1.5 STANDALONE substantive substrate framework support.

7. Closure

The substrate-primary Pochhammer primitive $3\pi/2^{C_2} = 3\pi/64$ appears in BOTH the lepton mass coefficient ($m_{e_substrate} / m_{anchor}$) AND the Yukawa coupling ($y_{e_substrate}$), as a substantive substrate-mechanism cross-link per Elie Toy 3695 + Toy 3709. The “+1 anomaly” identity $g - 1 = C_2$ provides the substrate-mechanism bridge from K-type Bergman norm ($1/2^g$) to mass-coupling primitive ($1/2^{C_2}$).

Casey-named #13 (Per-Generation Cluster Independence) gen-1 lepton substrate cluster $\{N_c, \pi, 2^{C_2}\}$ EXPLICITLY identified.

K3 v0.4 RS-coding mechanism + K-type dimensions v0.1 + Pochhammer primitive cross-link together form a substantive substrate-mechanism foundation for the m_e numerical closure (the remaining OPEN K3 element). Multi-week closure path identified.

Per Cal #35-honest: ONE Pochhammer primitive, TWO physical interpretations. Strong-Uniqueness v1.5 STANDALONE 10 legs unchanged.

— Lyra, Tue 2026-06-02 ~11:42 EDT. Substrate-Primary Pochhammer Mass-Yukawa Cross-Link v0.1: substrate-mechanism convergence on $3\pi/2^{C_2}$ primitive across mass + Yukawa observables; Casey #13 gen-1 cluster strengthened; cross-link to K3 v0.4 RS-coding via C_2 substrate-primary anchor.